Lecture 6: Predict-then-Optimize

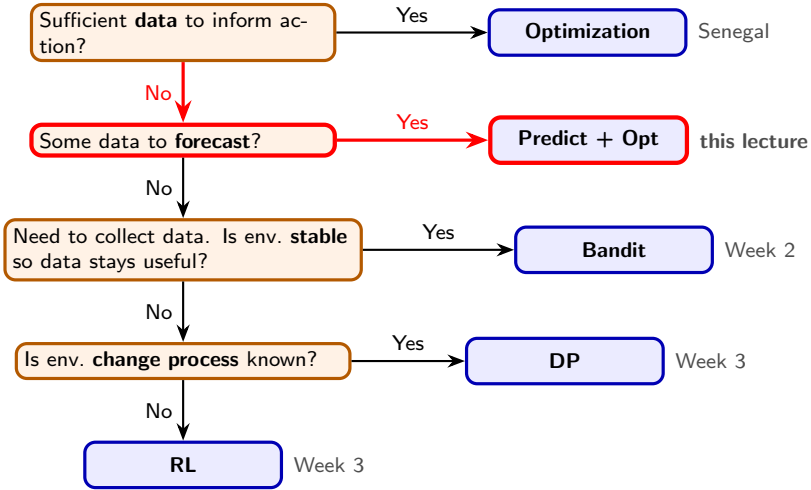
Forecast the Unknown Parameters, Then Solve the Problem

Peter Zhang

12768/19867/94867: From Data to Action

1. The Predict-then-Optimize Pattern
2. Case 1: Day-Ahead Energy Markets
3. Case 2: Uber Eats Push Notifications
4. Case 3: Federal Block Grants
5. What the Three Cases Share

A Very Likely Data-to-Action Path in Policy Analytics



Federal Block Grants: Predict + Optimize

Context: $1.5B+ in federal block grants each year for substance use disorder (SUD) treatment.

- **Prediction:** States forecast overdose mortality rates. Example: State A and State B similar population sizes. State A predicts 20 deaths / 100k vs. State B 40 deaths / 100k.
- **Optimization:** Allocate limited funds across 2 states.
- **Numerical example:** If $100M is available, - allocating by population alone might send $50M to State A and $50M to State B, - but allocation weighted by forecasted need might shift this to $30M vs. $70M.

Forecasting need changes the optimal allocation.

Incentive misalignment? Use independent epidemiological data, third-party audits, or formulas blending population, past outcomes, and so on.

Day-Ahead Energy Markets: Predict + Optimize

Context: Electricity markets clear *one day in advance*, scheduling billions of dollars of generation.

- **Prediction:** Tomorrow's demand + renewable output. Example: Forecast 100 GW peak demand, 30 GW wind.
- **Optimization:** Dispatch thermal plants, storage, and reserves.
- **Numerical example:** If demand is underestimated by 5 GW, reserve plants may need to start at \$300/MWh (compare with how much you pay). Accurate forecasts save tens of millions in balancing costs.

Forecasts drive tomorrow's dispatch and prices. Errors are absorbed by the market.

FEMA Disaster Response: Predict + Optimize

Context: FEMA must position supplies and teams *before* disasters strike.

- **Prediction:** Forecast hurricane path or flood zones. Example: 70% chance landfall in Florida, 30% in Georgia.
- **Optimization:** Pre-position water, food, medical kits, and trucks.
- **Numerical example:** If 500 trucks are available, 350 may be staged in Florida, 150 in Georgia. Wrong forecasts mean wasted effort and delayed relief.

Better predictions = faster and fairer relief.

Health Operations Planning: Predict + Optimize

Context: Hospitals and cities plan for surges from disasters or epidemics (recall Week 1 SIR example, and Assignment 1).

- **Prediction:** Model storm surge, wildfire spread, or epidemic curves. Example: SIR model predicts 500 ICU patients in 2 weeks.
- **Optimization:** Allocate staff, ventilators, and open surge wards.
- **Numerical example:** If a city has 300 ICU beds, planning based on forecasts means activating overflow sites early.

Forecasts buy time for optimal deployment of scarce resources.

Recall the MIP Model in A1

- Given Parameters

Staffing demand	$\text{demand}_{d,s} \in \mathbb{Z}_{\geq 0}$
Full-time employee availability	$\text{availability}_{e,d} \in \{0, 1\}$

- Decision Variables

Full-time staffing	$x_{e,d,s} \in \{0, 1\}$
Part-time hiring	$y_{d,s} \in \mathbb{Z}_{\geq 0}$

- Objective Function

$$\min \sum_{d,s} y_{d,s}$$

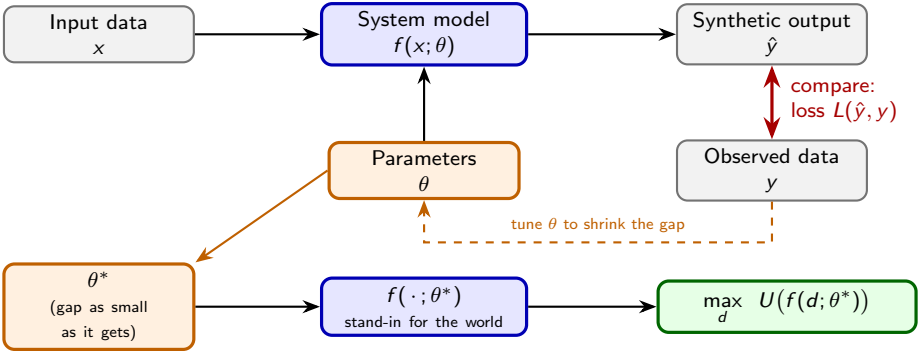
- Constraints

$$\sum_{e} x_{e,d,s} + y_{d,s} \geq \text{demand}_{d,s} \quad \forall d \quad \forall s$$

...

Predict-then-Optimize, in One Picture

Predict the unknown parameters of an optimization problem, then solve the problem using the predicted values.



1 The Predict-then-Optimize Pattern
2 Case 1: Day-Ahead Energy Markets
3 Case 2: Uber Eats Push Notifications
4 Case 3: Federal Block Grants
5 What the Three Cases Share

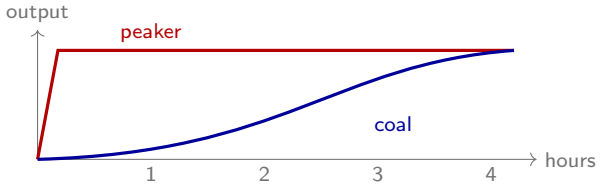
Day-Ahead Energy Markets

Basic Idea: Electricity markets schedule supply one day in advance to keep the grid balanced.

- **Why "day-ahead"**: Generators need at least hours to ramp up or shut down.
- **Clearing**: Matching predicted demand with generation offers, deciding *who produces, how much, and at what price*.
- **Who is involved?** - Grid operator (e.g., PJM in mid-Atlantic U.S.). - Generators (coal, gas, nuclear, wind, solar, hydro). - Large industrial consumers, and retailers who resell to households.

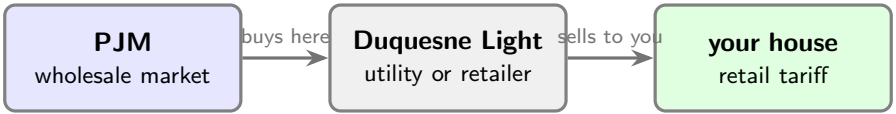
Who Can Move, and How Fast

Type	Time to start	Rough $/MWh	What it is for
Nuclear	days	very low fuel	runs flat out, almost never moves
Coal	6 to 12 hours	moderate	planned days ahead
Combined-cycle gas	1 to 3 hours	moderate	the workhorse of the fleet
Gas peaker	10 to 30 min	high	the hour you did not plan for
Battery	seconds	varies	fast, but only for a few hours
Wind and solar	n/a	near zero fuel	you take it when it is there



Figures are rough industry ranges, for a sense of scale.

How It Reaches Your House



- Only the first arrow is a market. Generators, large industrial consumers and retailers bid there. Households do not.
- Your utility buys wholesale, often through periodic procurement auctions, and sells to you at a regulated tariff that is reset on a slow schedule.
- So a $300 hour does not land on your bill that evening. It arrives later, averaged in, and nobody can point at it.

Decision Sequence and Market Size

Timeline of Decisions:

1. **Day-ahead (DA):** Forecast demand, clear market, schedule plants.
   (Long-term contracts are settled separately. We are not touching those.)
2. **Real-time (RT):** Adjust to actual demand, pay reserves or curtail.
3. **Settlement:** Prices and payments flow to generators and retailers.

Scale of money:

- PJM's day-ahead market: \$40B+ annually.
- U.S. wholesale markets: \$400B+.
- Europe, Asia run similar DA markets (Nord Pool, EPEX, etc.).

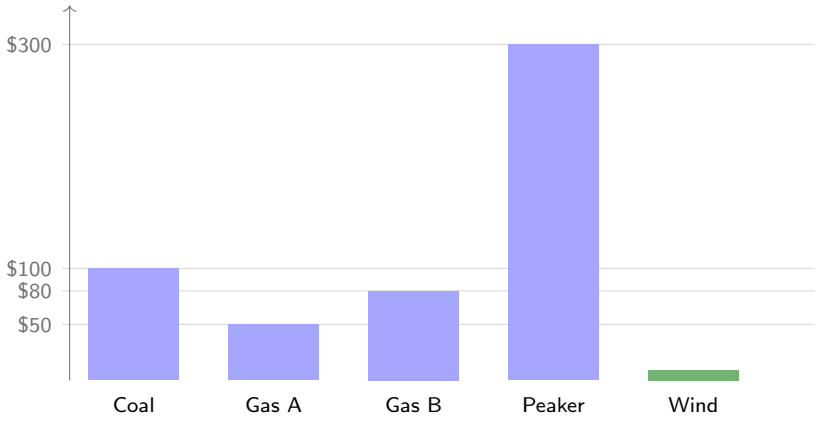
Billions of dollars hinge on tomorrow's forecast and optimization.

CASE 1 · PART 1 OF 3

The picture

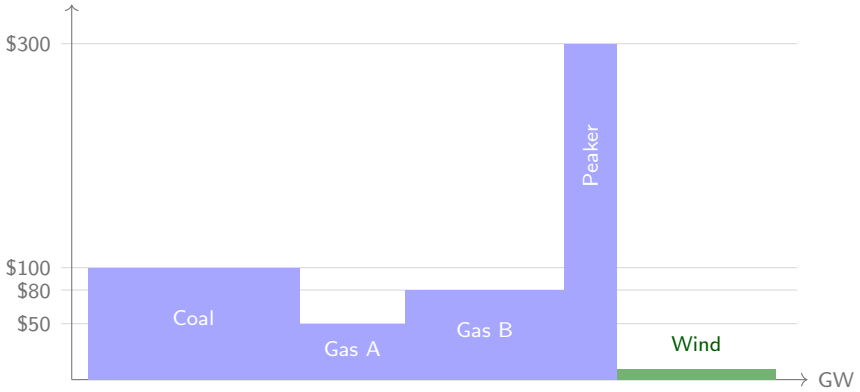
let us draw it

What Each Generator Charges



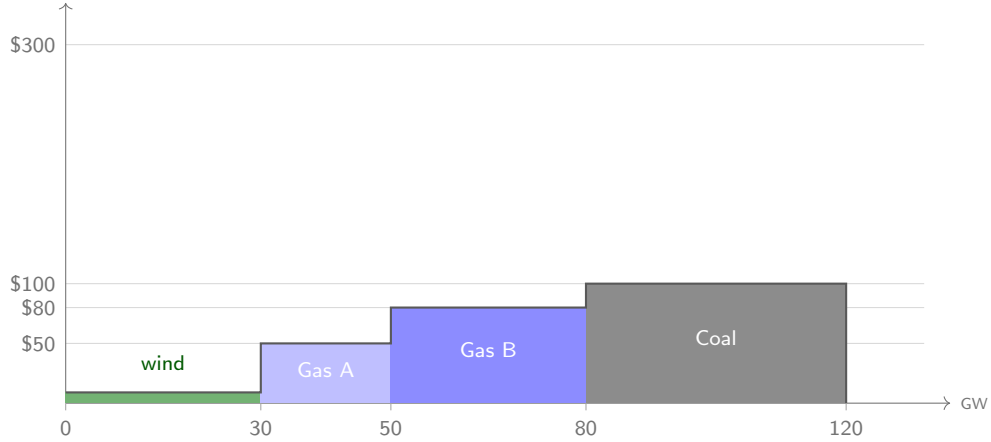
Five generators, and what each one wants per megawatt-hour.

They Are Not the Same Size



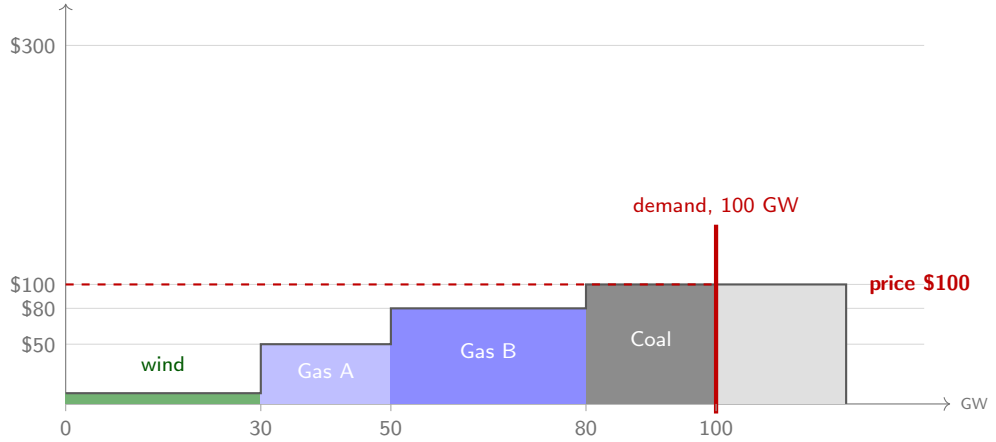
Now the width is how much each one can make.

The Merit Order



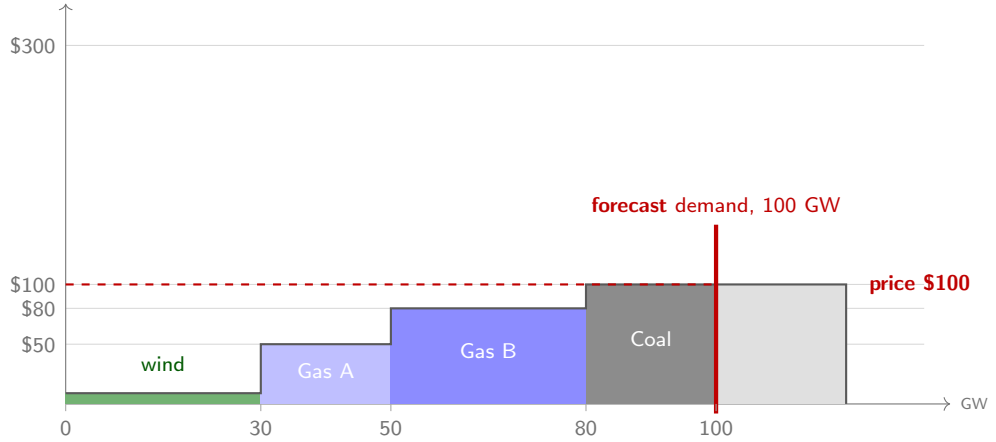
Same five generators, now lined up cheapest first. This ordering is called the **merit order**.

Where the Price Comes From



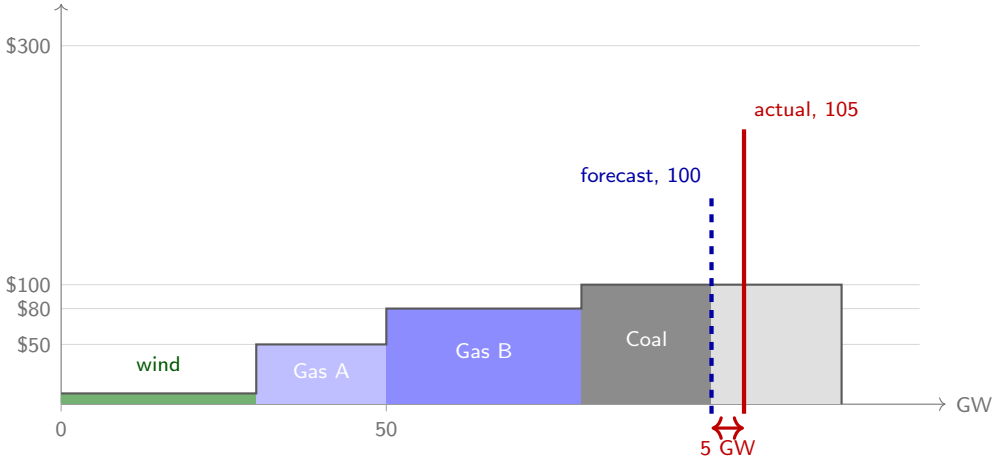
The **marginal** generator, the last one you need, sets the price. Everybody is paid that same **clearing price**, including the wind farm whose fuel was free.

One Word Changes Everything



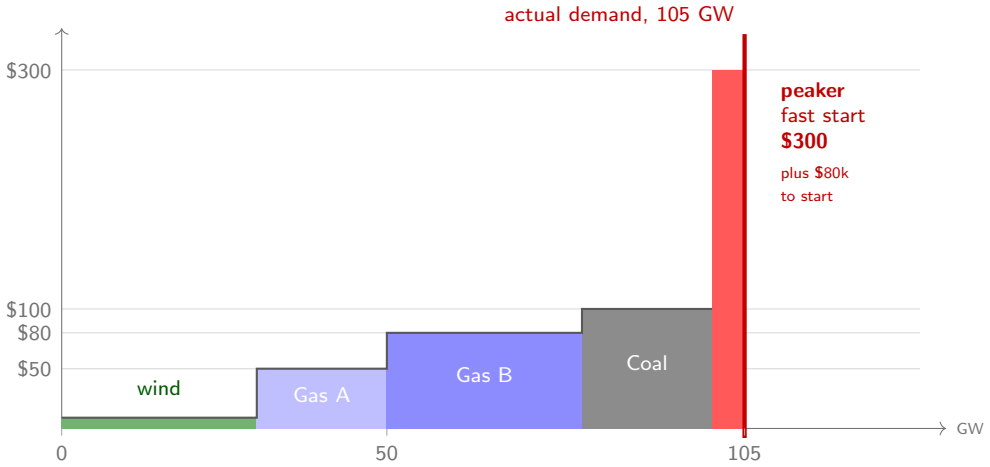
That vertical line is a **forecast**. Tomorrow has not happened yet, and everything to its left was committed on the strength of a prediction.

The Forecast Was 100. The Day Was 105.



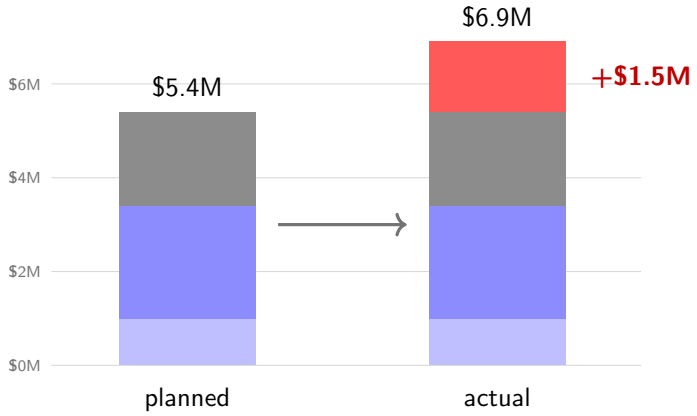
Five gigawatts, on a hundred. Everything that follows comes from that gap.

Actual Demand Is 105 GW



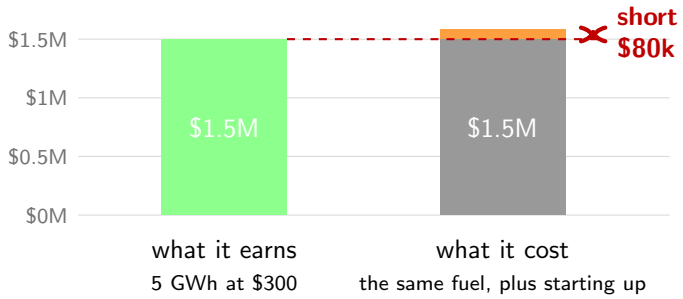
A **peaker**: an aeroderivative gas turbine (a jet engine on the ground), a reciprocating-engine plant, hydro, or increasingly a battery. Minutes to full output, and priced accordingly.

Cost of the Hour, Planned and Actual



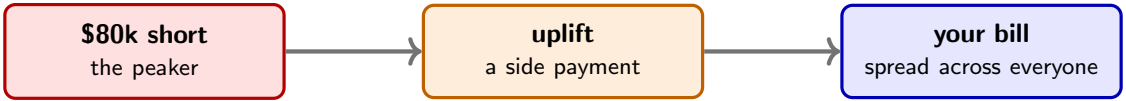
A **5%** error in demand produced a **28%** increase in cost.

The Peaker's Revenue and Cost



As the last unit needed, the peaker sets the price, so it is paid exactly its fuel cost. Its margin is **zero**, and the $80k start-up cost is not recovered.

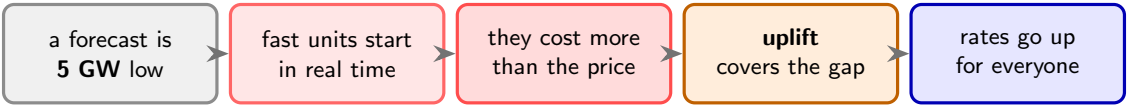
Uplift Cost Trickles Down



PJM pays out **tens to hundreds of millions** of dollars of this a year.

This does not appear in any forecast accuracy report.

From Forecast Error to Electricity Rates



A forecast error becomes a cost through a chain that nobody forecasts.

What If the Forecast Is Too High?

	Forecast too LOW	**Forecast too HIGH**
What happens	Start fast expensive units in real time	Too much generation already committed and running
Why it is not free	\$300 energy, plus start-up costs nobody recovers	Units have **minimum run levels**. You cannot simply switch them off.
Who absorbs it	Reserves and uplift	Wind gets curtailed even though its fuel is free, and prices can go **negative**

- Negative prices are real. A generator will pay to keep producing when shutting down and restarting costs more than the loss, or when a production subsidy depends on output.
- Wind forecast errors do the same thing from the other side. Over-forecast the wind and you are short of energy in real time. Under-forecast it and you have committed thermal plant you no longer need.

Errors in both directions cost money. They just cost it through different mechanisms.

CASE 1 · PART 2 OF 3

The numbers

walking through some numbers

The Data

Unit	Capacity	Offer	Cost to start
Wind	30 GW	\$0	
Gas A	20 GW	\$50	
Gas B	30 GW	\$80	
Coal	40 GW	\$100	already on
Peaker *(fast start)*	10 GW	\$300	**\$80k**

Forecast demand for tomorrow's peak hour: **100 GW**

The same information as the merit order figure.

Step 1. Which Units Run

Fill demand starting from the cheapest, and stop when you reach 100 GW.

Unit	**Take**	**Running total**	
Wind	30 GW	30 GW	all of it
Gas A	20 GW	50 GW	all of it
Gas B	30 GW	80 GW	all of it
Coal	20 GW	**100 GW**	**only half of its 40 GW**
Peaker	0		not needed

Coal is the **last unit needed**.

Step 2. The Price and the Cost

The price is the offer of the last unit we needed. Coal. So **$100/MWh**, and every generator is paid that, including the wind.

	Energy	**Its cost**	**Total**
Wind	30 GWh	$0	$0
Gas A	20 GWh	$50	$1.0M
Gas B	30 GWh	$80	$2.4M
Coal	20 GWh	$100	$2.0M
	cost of the hour		**$5.4M**

Step 3. Actual Demand Is 105

Coal was committed at 20 GW. It has 20 GW spare, and it cannot get there in five minutes. Only the peaker can.

Planned cost of the hour	\$5.4M
Peaker, 5 GWh at \$300	**+\$1.5M**
Actual cost of the hour	**\$6.9M**

The demand forecast was off by **5%**, and the cost by **28%**.

Step 4. The Peaker's Revenue and Cost

The peaker was the last unit needed, so it is **marginal**, and the price becomes its offer: **$300**. A generator offers its fuel cost, so it earns back exactly what the fuel cost.

It earns	5 GWh at $300	$1,500k
Fuel	5 GWh at $300	$1,500k
	margin	**$0**
Starting up	one-off	$80k
	short by	**$80k**

The marginal unit earns zero margin, so a per-megawatt-hour price cannot recover any **fixed** cost it incurred.

The operator covers the $80k. That payment is **uplift**.

CASE 1 · PART 3 OF 3

The general form

now in general

Ingredients

Sets and actors

- Generators $g \in \mathcal{G}$ offer power at price c_g (\$/MWh).
- Loads $d \in \mathcal{D}$ have fixed demand q_d (MW).
- Transmission lines $\ell \in \mathcal{L}$ carry limited flow.

Decision variables

- p_g: how much each generator produces.
- f_ℓ: how much flow on each line.

The market decides p_g and f_ℓ to meet demand at lowest cost.

Day-Ahead Market: Optimization Problem

Objective: minimize cost of generation

$$\min_{p,f} \quad \sum_{g \in \mathcal{G}} c_g \, p_g$$

Constraints:

$$\begin{aligned} \text{(Balance)} \quad & \sum_g p_g = \sum_d q_d \\ \text{(Line limits)} \quad & -\overline{F}_\ell \leq f_\ell \leq \overline{F}_\ell \\ \text{(Generator limits)} \quad & 0 \leq p_g \leq \overline{P}_g \end{aligned}$$

Interpretation:

- Dispatch cheapest plants first.
- Always keep supply = demand.
- Respect line capacities.

Dual Variables (Shadow Prices)

Duals give economic meaning:

- λ: value of one more MW of demand (system marginal price).
- μ_ℓ^+, μ_ℓ^-: value of relaxing a line limit (congestion prices).
- α_g^+, α_g^-: value of relaxing generator capacity (scarcity rents).

Stationarity intuition: At optimum, each generator is indifferent:

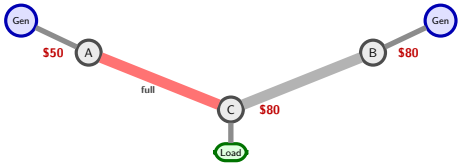
$$\text{Offer price } c_g = \text{Price at its bus (LMP)}.$$

(Bus: the connecting node between generator and grid.)

This is why clearing prices emerge "for free" from optimization.

Locational Marginal Prices (LMPs)

$$\mathrm{LMP}_n = \underbrace{\lambda}_{\text{system}} + \underbrace{\sum_{\ell} (\mu_\ell^+ - \mu_\ell^-)\,\mathrm{PTDF}_{\ell n}}_{\text{congestion}}$$



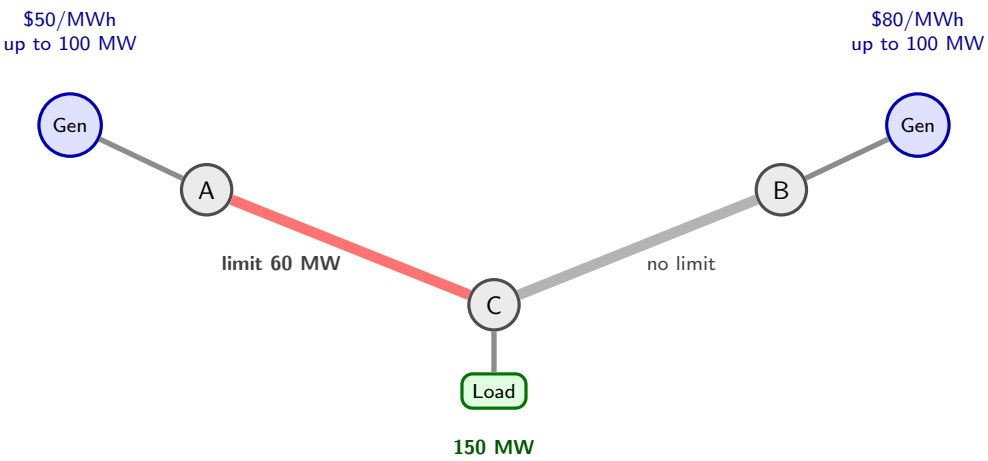
Same instant. Three buses. Two prices.

- Loads at n pay LMP_n. Generators at n are paid LMP_n.
- With no congestion, every $\mu = 0$ and one price covers the whole system.
- A binding line makes prices **diverge**, by exactly that line's shadow price.

PTDF, power transfer distribution factor: if you inject one extra MW at bus n and withdraw it at the reference bus, $\mathrm{PTDF}_{\ell n}$ is the share of that MW that physically flows down line ℓ. Electricity splits across every available path, so this is rarely 0 or 1.

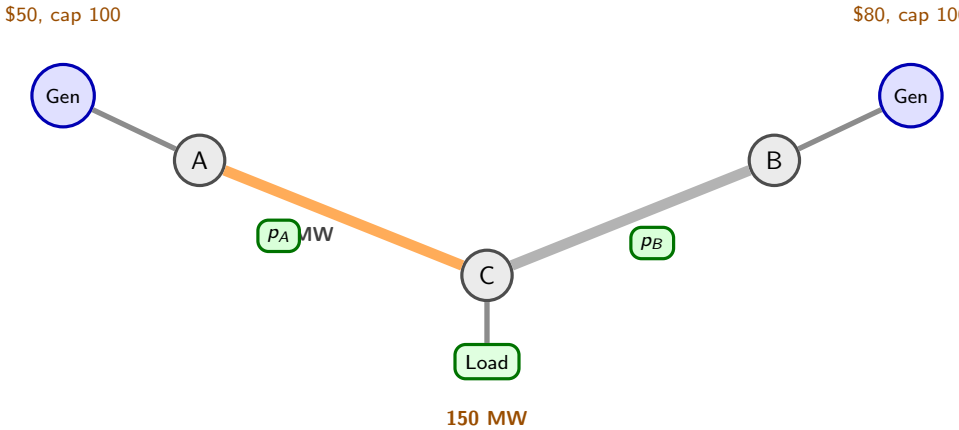
Where a generator sits relative to its own limits: if $0 < p_g < \overline{P}_g$ it is marginal at its bus and $c_g = \mathrm{LMP}_{n(g)}$; if $p_g = \overline{P}_g$ it is capped and $\mathrm{LMP}_{n(g)} > c_g$, which is where its profit comes from; if $p_g = 0$ it was too expensive and $\mathrm{LMP}_{n(g)} < c_g$.

A Three-Bus Network



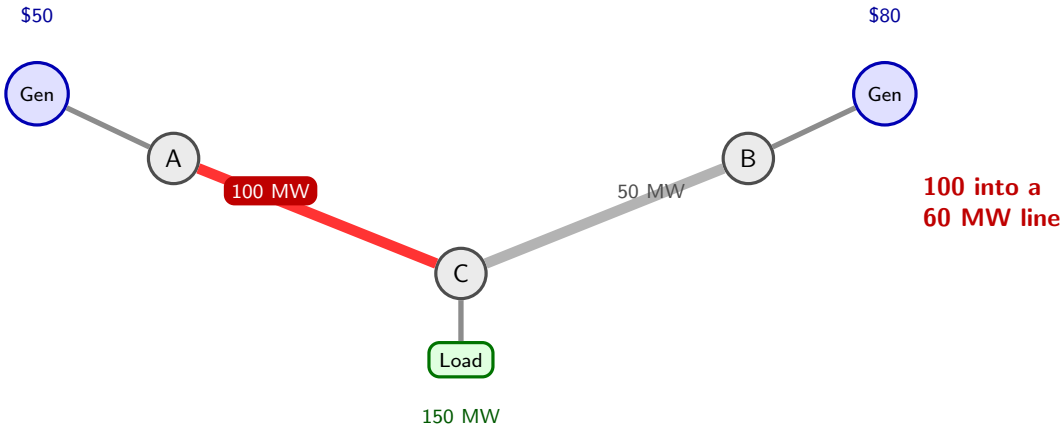
The cheaper generator is separated from the load by a line with limited capacity.

Parameters and Decision Variables



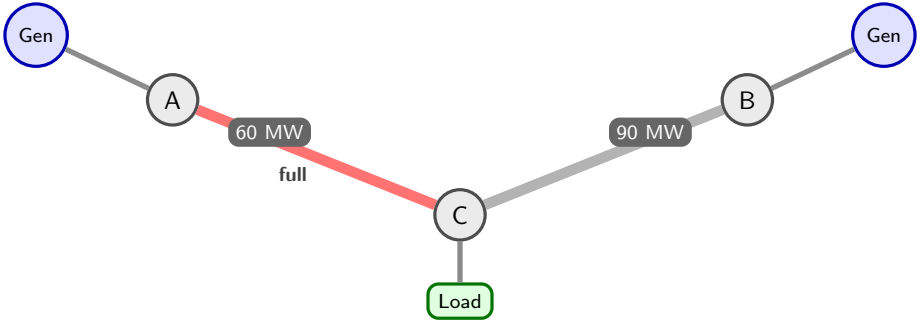
Orange values are given. **Green** values are chosen.

Cheapest-First Violates the Line Limit



Cheapest-first would cost $9,000, but it puts 100 MW on a 60 MW line.

The Feasible Dispatch



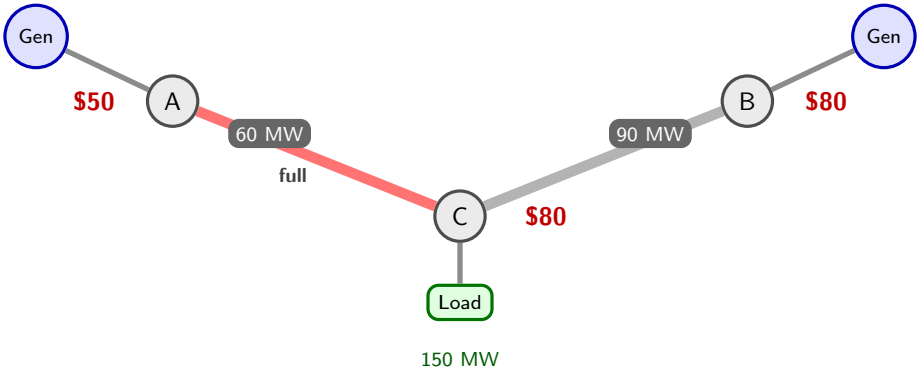
Gen A	60 MWh at \$50	\$3,000
Gen B	90 MWh at \$80	\$7,200
	total	**\$10,200**

The Shadow Price of the Line

	Gen A	Gen B	Cost
Line holds 60	60	90	\$10,200
Line holds 61	61	89	\$10,170
		saving	**\$30**

\$30 is the **shadow price** of the line: the value of one more megawatt of capacity.

Prices at the Three Buses



One more MW at **A** comes from A, so it costs \$50.

One more MW at **C** has to come from B, so it costs \$80.

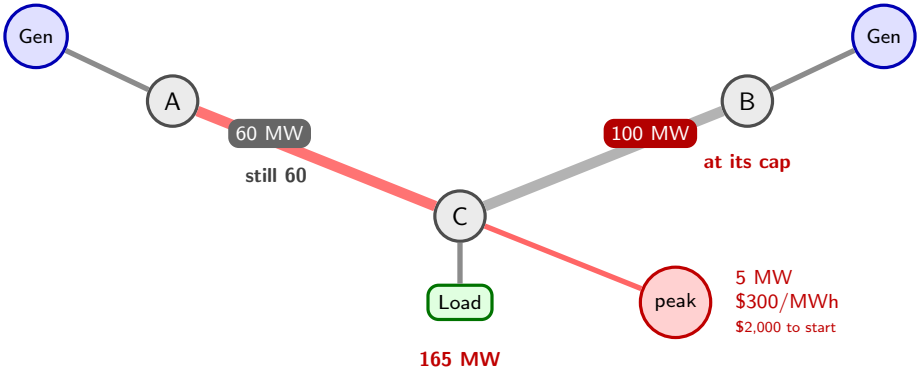
The \$30 difference across the line equals the shadow price from the previous slide.

Payments and Congestion Rent

Load at C pays	150 MWh at \$80	\$12,000
Gen A is paid	60 MWh at \$50	−\$3,000
Gen B is paid	90 MWh at \$80	−\$7,200
	left over	**\$1,800**

The operator collects \$1,800 more than it pays out. This is **congestion rent**, equal to 60 MW times the \$30 price difference.

Actual Demand Is 165



Gen A is limited by the line and Gen B is at its capacity, so the remaining 5 MW comes from the peaker.

Prices After the Peaker Starts

	Runs	Its cost	Paid at C price $300
Gen A	60 MWh	$50	$50, still stuck behind the line
Gen B	100 MWh	$80	$300
Peaker	5 MWh	$300	$300, the last unit needed

Gen B earns $30,000 on fuel that cost $8,000, because it sits on the unconstrained side of the network.

The Peaker's Revenue and Cost, on the Network

It earns	5 MWh at \$300	\$1,500
Fuel	5 MWh at \$300	\$1,500
	margin	**\$0**
Starting up	one-off	\$2,000
	short by	**\$2,000**

The last unit needed earns zero margin, so its start-up cost becomes **uplift**. The same mechanism as before, now on a network.

How Prediction Moves Prices (and How Errors Misprice)

Forecast ⇒ Optimize ⇒ Prices.

- **Prediction sets parameters** in the DA LP: $\{\widehat{q}_d,\ \widehat{\text{wind/solar}},\ \overline{F}_\ell^{\text{DA}}\}$.
- **Clearing solves** the LP and yields duals $(\lambda^{\text{DA}}, \mu^{\text{DA}})$,

$$\text{LMP}_n^{\text{DA}} = \lambda^{\text{DA}} + \sum_\ell (\mu_\ell^{\text{DA}+} - \mu_\ell^{\text{DA}-})\,\text{PTDF}_{\ell n}.$$

When predictions are wrong (RT):

- Realizations $\{q_d^{\text{RT}},\ \text{VRE}^{\text{RT}},\ \overline{F}_\ell^{\text{RT}}\}$ change which constraints bind ⇒ new duals $(\lambda^{\text{RT}}, \mu^{\text{RT}})$,

$$\text{LMP}_n^{\text{RT}} = \lambda^{\text{RT}} + \sum_\ell (\mu_\ell^{\text{RT}+} - \mu_\ell^{\text{RT}-})\,\text{PTDF}_{\ell n}.$$

- **Price error at bus n:** $\Delta\text{LMP}_n = \text{LMP}_n^{\text{RT}} - \text{LMP}_n^{\text{DA}}$.

Who Decides What: ISO, Generators, Loads

Inputs to ISO/RTO

- **Generators:** curves $c_g(\cdot)$, start-up/min-load, ramp, reserves, bus location.
- **Loads/retailers:** demand bids (fixed or price-responsive), virtual bids.
- **ISO:** forecasts (load, wind/solar), network limits $\overline{F}_\ell$, reserve reqs.

What the ISO controls

- **Quantities**: on/off commitment, dispatch p_g, flows f_ℓ, reserves.
- **Not prices**: *prices emerge* as dual (shadow) prices of the dispatch problem.

What is solved (conceptually)

- **SCUC** (unit commitment, MIP): on/off with start-up/min up-down, ramping.
- **SCED** (economic dispatch, LP/QP): nodal balance + transmission limits (DC-OPF).
- *LMPs come from SCED duals* (duals from SCUC not for pricing).

Timeline & settlement

- Gate closure $\rightarrow$ SCUC/SCED $\rightarrow$ publish DA schedules & **DA LMPs**.
- Real-time (RT) re-dispatch $\rightarrow$ **RT LMPs**.
- Settlement: DA schedules at DA LMPs; deviations at RT LMPs; uplift/make-whole for non-convex costs.

Takeaway: *ISO chooses quantities; prices are duals. Predictions change what binds, so they change prices.*

For Reference: Primal, Dual, LMP

Primal (one hour SCED)

$$\min_{p,f} \sum_g c_g\, p_g \quad \text{s.t.} \quad \underbrace{\sum_g p_g = \sum_d q_d}_{\text{system/bus balance}},\ \underbrace{-\overline{F}_\ell \le f_\ell \le \overline{F}_\ell}_{\text{line limits}},\ \underbrace{0 \le p_g \le \overline{P}_g}_{\text{gen limits}}.$$

Key duals (shadow prices)

λ_n (balance at bus n) $\qquad \mu_\ell^+, \mu_\ell^- \ge 0$ (line ℓ upper/lower) $\qquad \alpha_g^+, \alpha_g^- \ge 0$ (gen upper/lower).

Locational Marginal Price (bus n)

$$\boxed{\mathrm{LMP}_n = \lambda_n + \sum_\ell (\mu_\ell^+ - \mu_\ell^-)\,\mathrm{PTDF}_{\ell n}}$$

(PTDF: flow sensitivity of line ℓ to 1 MW at n).

Who pays/receives Loads at n pay LMP_n; generators at n receive LMP_n.

Generator KKT (why α matters)

$$0 < p_g < \overline{P}_g \Rightarrow \alpha_g^{\pm} = 0 \Rightarrow c_g = \mathrm{LMP}_{n(g)} \text{ (on the margin)};$$

$$p_g = \overline{P}_g \Rightarrow \alpha_g^+ > 0 \Rightarrow \mathrm{LMP}_{n(g)} < c_g; \quad p_g = 0 \Rightarrow \alpha_g^- > 0 \Rightarrow \mathrm{LMP}_{n(g)} > c_g.$$

Forecasts $\Rightarrow$ prices Changing $\widehat{q}_d$, VRE, or $\overline{F}_\ell$ changes which constraints bind $\Rightarrow$ different (λ, μ) $\Rightarrow$ different LMPs.

Doing This in Code

1. PREDICT, the usual way

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

Any fitted scikit-learn regressor works.

2. OPTIMIZE, with the model inside

```
import gurobipy as gp
from gurobi_ml import add_predictor_constr

m = gp.Model()
x    = m.addMVar(n_features)    # decision
yhat = m.addMVar(1)             # prediction

add_predictor_constr(m, model, x, yhat)

m.setObjective(yhat, gp.GRB.MAXIMIZE)
m.optimize()
```

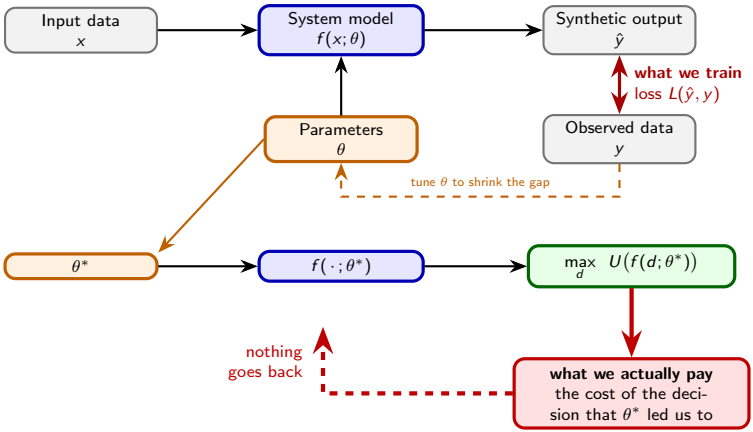
`add_predictor_constr` rewrites the fitted model as constraints the solver can see, so the optimizer can choose x knowing what the model would predict. Linear and logistic regression, decision trees, random forests, gradient boosting and small neural networks are all supported.

The Energy Market in Method Terms

Predicted	tomorrow's demand, and wind and solar output
Optimized	which generators run, and how much each produces
Decided when	roughly a day before the electricity is actually needed
Error shows up as	reserves fired at \$300/MWh, and uplift charges

A forecast error is expensive in proportion to how much it changes the dispatch decision, which is only loosely related to its size in an accuracy report.

What We Train vs What We Actually Pay



The training loop only ever sees the **loss**. The organisation only ever feels the **decision cost**. Nothing connects the second back to θ.

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Uber Eats Push Notifications

A commercial setting: how to optimize user engagement and conversion using the predict-then-optimize framework.

Uber Eats Push Notifications

Uber introduced a system to manage push notifications from various teams on a user level, where **all incoming pushes are persisted and buffered**

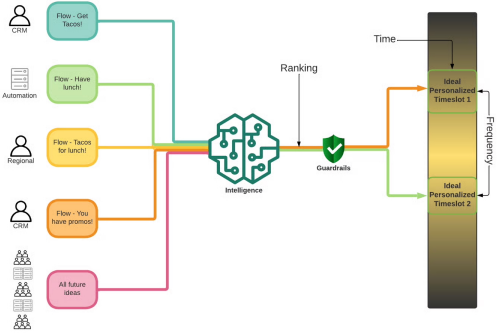


Image Credit: Vinay Sharma et al.

Vinay Sharma, Rémi Torracinta, Giacomo Lamberti, & Britton Overall. (2022, November). How Uber Optimizes the Timing of Push Notifications using ML and Linear Programming. Retrieved August 1, 2025, from https://www.uber.com/en-EG/blog/how-uber-optimizes-push-notifications-using-ml/

The Business Question

What is the best schedule over a time horizon with which to send the pushes currently buffered for a user?

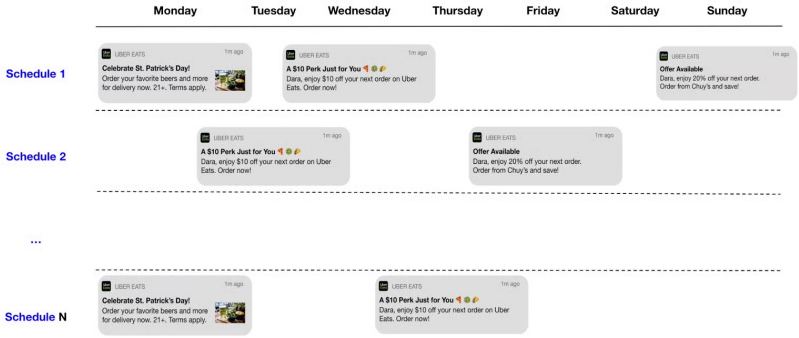


Image Credit: Vinay Sharma et al.

From Business to Technique

What is a "best schedule"?

A schedule with

- quickest notification click
- highest order payment
- optimal frequency
- balanced timing, avoiding fatigue or annoyance
- sustained engagement, e.g., long-term user retention and satisfaction
- **conversion rate**

Predict-Then-Optimize Framework

- Given a set of candidate pushes $i \in I$ for a user u and a set of possible delivery times $t \in T$
- **XGBoost predicts** the probability of a user u making an order within 24 hours of receiving push i at time t

$$s_{i,t} = p(order_{24h}|u, i, t)$$

- **MIP decides** whether push i to send at time t,

$$x_{i,t} \in \{0, 1\}$$

maximizes the total order probability for user u

$$\max \quad \sum_{i,t} x_{i,t} \; s_{i,t}$$

Full Problem Formulation

With	$s_{i,t}$	order probability of assigning push i to time t
$\max_{x_{i,t}}$	$\sum_{i,t} x_{i,t} s_{i,t}$	the best schedule with the highest probabilities
subject to	$x_{i,t} \in \{0,1\}$,	send push i at time t, or don't
	$\sum_{t} x_{i,t} \leq 1, \ \forall i$,	send a push at most once
	$\sum_{i} x_{i,t} \leq 1, \ \forall t$,	send at most one push at a time
	$\sum_{i,t} x_{i,t} \leq F$,	send at most F pushes in the time horizon
	$x_{i,t} \leq \begin{cases} 1 & \text{if } t < expiration_i \\ 0 & \text{if } t \geq expiration_i \end{cases}$,	don't send a push after expiration
	$x_{i,t} \leq \begin{cases} 1 & \text{if } t \in window_i \\ 0 & \text{if } t \notin window_i \end{cases}$,	don't send a push outside of its windows

A Refined, More Complex Variant

What if conversion rates are NOT independent across pushes?

- Assume conversion rates also depend on push decisions of previous 2 time steps, decisions would influence both prediction and optimization
- **Prediction model**

$$s_{i,t} = p\left(order_{24h} \middle| u,\ i,\ t,\ \left\{ x_{j,t'} : j \in I,\ t' \in \{t-2, t-1\} \right\} \right)$$

- **Optimization objective function**

$$\max \quad \sum_{i,t} x_{i,t}\ s_{i,t}$$

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SUD Treatment Center Allocation

A high-stakes public health setting: how to allocate scarce treatment resources while balancing efficiency and fairness.

SUD Treatment Center Allocation

Public health challenge: Substance Use Disorder (SUD) is a major cause of overdose deaths in the U.S.

- 18,000+ treatment centers nationwide, but mismatched with overdose mortality
- Federal block grants: \$1.5B annually, 40% of public SUD treatment spending
- Policymakers face tradeoffs between efficiency and fairness

Source: see paper posted on Canvas under this week's module.

The Policy Question

How should states allocate limited SUD treatment centers across counties?

- **Efficiency:** Reduce overdose deaths
- **Equity:** Allocate relative to counties' overdose death burden
- **Equality:** Allocate relative to counties' populations
- Key challenge: today's allocations change tomorrow's death rates

Predict Stage: Treatment Effect Estimation

Step 1: Estimate how centers affect overdose deaths

$$\log(y_{t,m}) = \theta\, x_{t-1,m} + \beta^\top c_{t,m} + \mu_t + \mu_m + \varepsilon_{t,m}$$

- $y_{t,m}$ = overdose deaths per 100,000 residents
- $x_{t-1,m}$ = number of treatment centers in prior year
- $c_{t,m}$ = other county characteristics (population, poverty, etc.)
- μ_t, μ_m = time and county fixed effects
- **Interpretation:** $\theta < 0$ means more centers $\Rightarrow$ fewer deaths

Optimize Stage: Objectives

Step 2: Decide where to allocate centers each year

- **Efficiency (mortality):**

$$J_t(x_t) = \frac{1}{M} \sum_{m=1}^{M} \hat{y}_{t+1,m}(x_{t,m})$$

  Average predicted deaths across counties.

- **Inequity (death-based fairness):**

$$Q_t^{eq}(x_t) = \frac{1}{M} \left\| \frac{x_{t,m}}{c_{t,m}} - \frac{\sum_j x_{t,j}}{\sum_j y_{t,j} c_{t,j}} \cdot y_{t,m} \right\|_1$$

  Do allocations match counties' share of deaths?

- **Inequality (population-based fairness):**

$$Q_t^{ql}(x_t) = \frac{1}{M} \left\| \frac{x_{t,m}}{c_{t,m}} - \frac{\sum_j x_{t,j}}{\sum_j c_{t,j}} \cdot c_{t,m} \right\|_2$$

  Do allocations match counties' share of population?

Joint Optimization Problem

Balance efficiency and fairness over H years

$$\min_{\{x_{t,m}\}} \frac{1}{H} \sum_{t=t_0}^{t_0+H-1} \Big[\alpha \cdot J_t(x_t) + (1-\alpha)\big(\lambda \cdot Q_t^{eq}(x_t) + (1-\lambda) \cdot Q_t^{ql}(x_t)\big)\Big]$$

- α: weight on saving lives vs. fairness
- λ: within fairness, balance equity (deaths) vs. equality (population)
- Constraints: budget B_t, operating and opening costs, minimum baseline centers L

Results at a Glance

- Econometric estimate: $\theta \approx -0.0043 \Rightarrow$ each new center reduces deaths by $\sim 0.43\%$
- Optimal allocations:
  - Up to **14% fewer deaths**
  - **50–80% less inequity/inequality**
- Tradeoffs (Pareto frontier):
  - High α: centers surge to large urban counties
  - Low α: resources spread to smaller underserved counties

Policy Implications

- Naïve rules (e.g., proportional increases, population-only) can worsen disparities
- Data-driven optimization gives states levers:
  - Urban-heavy states: prioritize death reduction
  - Rural-heavy states: prioritize equity
- Explicit tradeoff makes policymaking more transparent

Managerial Insight

Optimization-based allocation can save lives *and* reduce disparities, but only if policymakers explicitly balance efficiency and fairness.

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What the Three Cases Share

	What is predicted	What is decided	Who pays for an error
Day-ahead market	demand and wind output tomorrow	which plants run, and how much	ratepayers, through reserves and uplift
Uber Eats	probability of an order within 24 hours, given user, push and time	which pushes go out, and when	lost orders, and users who mute the app
Block grants	effect of one more treatment center on overdose deaths	how many centers each county gets	counties left short, in lives

- In all three, what the organisation needs is the dispatch schedule, the notification plan, or the allocation. The forecast is an input to producing it.
- The forecast is an input to a decision. It gets trained by a loss function, and that loss function knows nothing whatsoever about the decision.

That gap is the entire subject of the next lecture. If the decision is what you care about, why are you training the forecast to minimise squared error?

Causal vs Non-causal

The SUD case uses econometric / causal models to do "prediction", while some other cases use purely machine learning (non-causal) models to do prediction. How do you decide what prediction model to use when you encounter a problem?

Answer: See the prediction policy problems paper on Canvas.